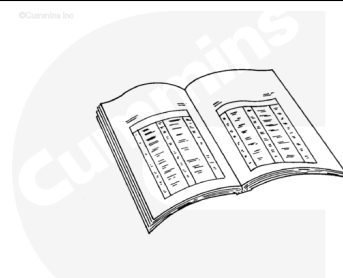


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

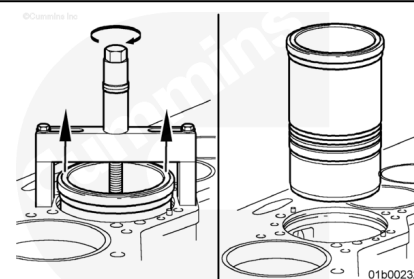
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Disconnect the batteries or air supply to the air starter to prevent accidental engine starting.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>)
- Drain the lubricating oil system and remove the oil pan. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)

- Remove the oil suction tube and the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/89/89-011-007.html)
- Remove the air intake manifold. Refer to Procedure 010-023 in Section 10.
(/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Remove the crossheads.
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html)
- Remove the connecting rod and piston assembly. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)

Remove

Use a liner puller, Part Number 3163745, and puller plate, Part Number 3163751, to remove the cylinder liner.

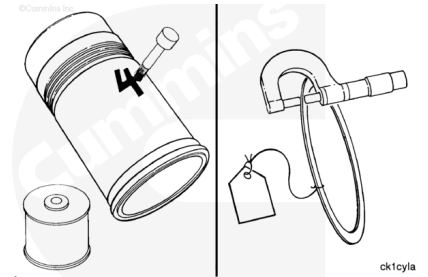


Use a liquid metal marker to mark the cylinder number and bank on each liner. Mark the cylinder liner on the camshaft side of the cylinder liner.

If sealing rings were used:

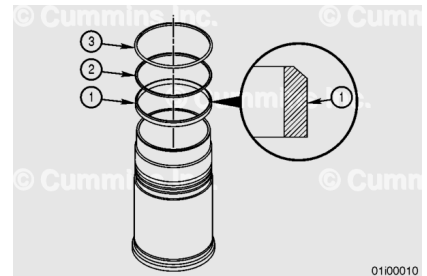
- Use a tag to mark the cylinder number.
- Measure in several places, and record the thickness of the sealing rings used in each cylinder.

The thickness of the sealing ring is one factor in determining liner protrusion. This information **must** be known when the liners are installed in the engine.



Remove the two D-rings (2) and (3).

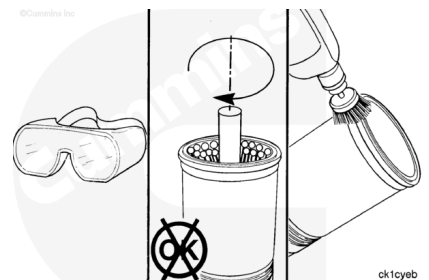
Remove and discard the crevice seal (1) .



Clean

⚠ WARNING ⚠

Wear eye protection. Make sure the wire brush is rated for the rpm being used if the brush is motor driven. Flying debris and dirt can cause personal injury.



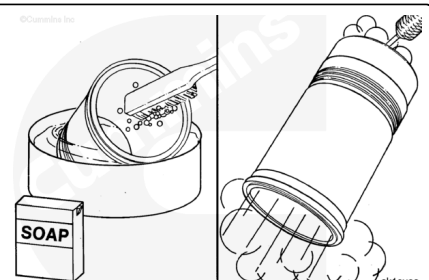
⚠ CAUTION ⚠

Do not use a hone, aluminum oxide cloth, sandpaper, deglazing, or pre-brushing to clean the cylinder liners. Abrasives can damage the finish and the crosshatch pattern and can contaminate the liner.

Use a high quality steel wire brush to clean the liner flange seating area and the secondary press fit below the liner flange.

⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

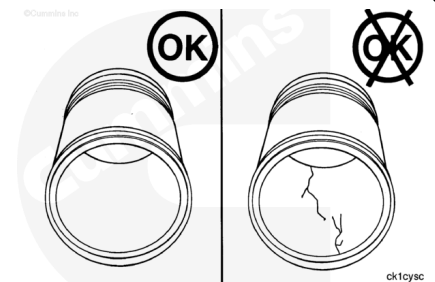
Do not contaminate the wash tank solution with bead blast materials. Use a kerosene emulsion-based solvent. Do not use a solvent that contains chlorinated hydrocarbons with cresol, phenols, or cresylic components. Engine damage can result.

Use a nonmetallic bristle brush, detergent soap, and warm water to clean the inside of the liner.

Clean the liners with steam or solvent.

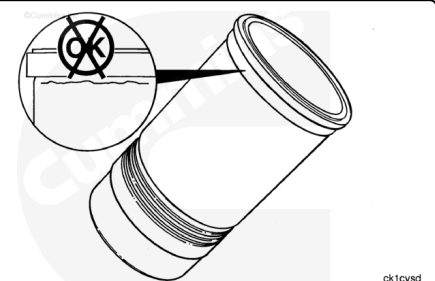
Dry with compressed air.

Inspect the liners for cracks on the inside and the outside surfaces.



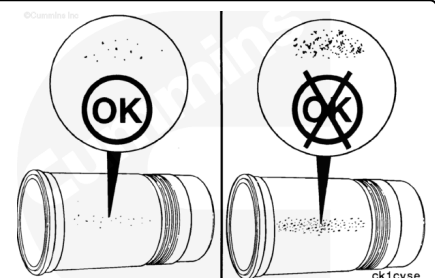
Inspect for cracks under the flange.

Cracks can also be detected by using either magnetic inspection or the dye method.



Inspect the outside diameter for excessive corrosion or pitting. Pits **must not** be more than 1.6 mm [0.063 in] deep.

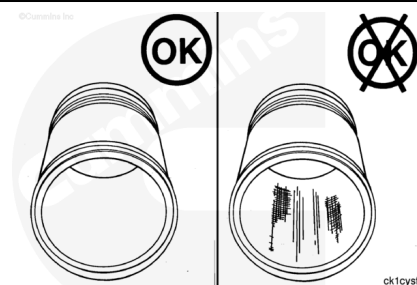
Replace the liner if the pits are too deep or if the corrosion can **not** be removed with a fine emery cloth.



Inspect the inside diameters for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the liner **must** be replaced.

Inspect the inside diameter for scuffing or scoring.

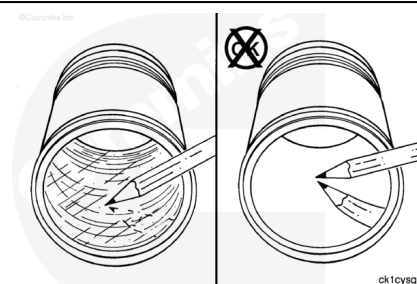


Inspect the inside diameter for liner bore polishing.

A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

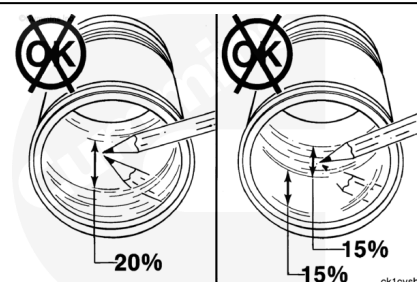
A heavy polish produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.

Reference Parts Reuse Guidelines, Bulletin 3810303, for further information on liner bore polishing.

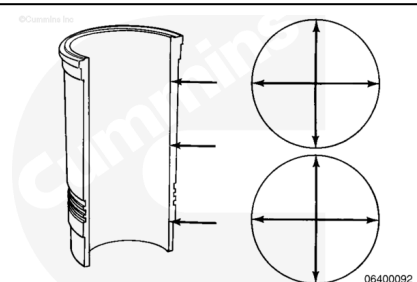


Replace the liner if:

- A heavy polish is present over twenty percent of the piston ring travel area.
- Thirty percent of the piston ring travel area has both moderate and heavy polish, with one-half of the 30 percent consisting of heavy polish.



Use a dial bore gauge to measure the inside diameter of the liner at the top, the bottom, and the middle of the piston ring travel area. Perform two measurements at each location. The measurements **must** be 90 degrees apart.



Cylinder Liner Inside Diameter

mm		in
170.040	MIN	6.694
170.240	MAX	6.702

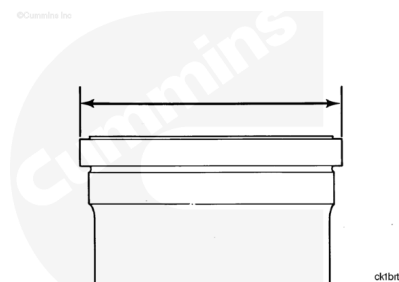
Measure the liner flange outside diameter.

Upper Press Fit Liner Flange Diameter

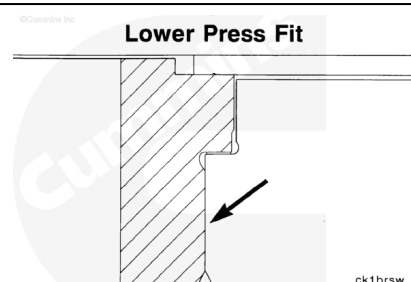
mm		in
205.965	MIN	8.109

Upper Press Fit Liner Flange Diameter

mm		in
206.015	MAX	8.111



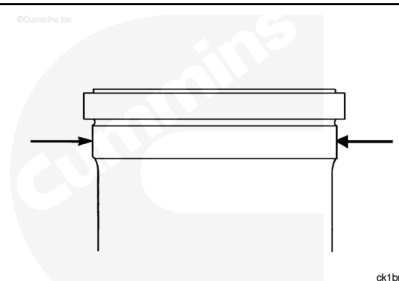
The liner design incorporates a press fit between the upper liner bore and the area of the liner directly below the liner flange. This is referred to as the lower press fit design.



Measure the lower press fit diameter.

Lower Press Fit Area Outside Diameter

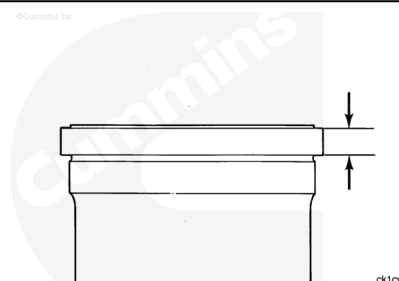
	mm		in
Standard Size	194.565	MIN	7.66
	194.615	MAX	7.662
0.25 mm [0.0098] Oversize	194.815	MIN	7.6699
	194.865	MAX	7.6719
0.50 mm [0.0197] Oversize	195.065	MIN	7.6797
	195.115	MAX	7.6817



Measure the liner flange thickness.

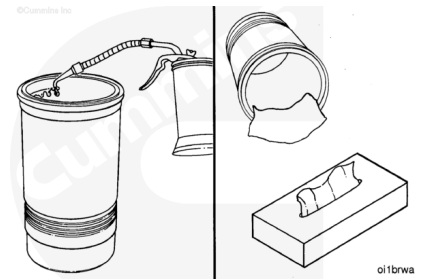
Liner Flange Thickness

mm		in
14.12	MIN	0.556
14.15	MAX	0.557



⚠ CAUTION ⚠

Do not place the liners in an area where dirty air flow can contaminate the liners. Engine damage can result.



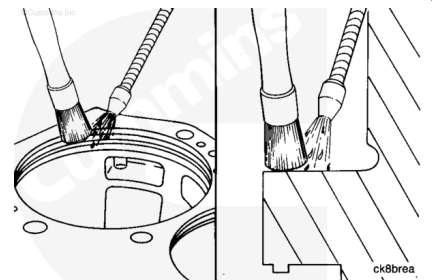
Apply a thick film of clean 15W-40 engine oil to the bores of the liners for final cleaning. Leave the oil on for 5 to 10 minutes.

Use a clean, lint-free paper towel to wipe the oil from the bores until the black and gray deposits are removed.

Install

⚠ WARNING ⚠

When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the bottom of the cylinder liner flange and the cylinder block upper and lower liner bore area with safety solvent.

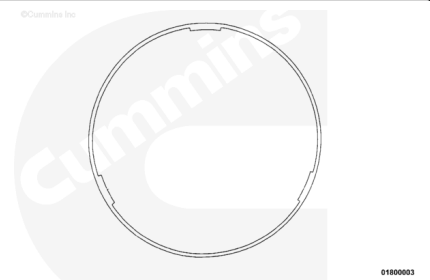
Note : Cylinder liner seal rings for the QSK23 are **not** installed at the factory. They are to be used at rebuild, after the machining process, to adjust liner protrusion.

The seal rings have three locating tabs on the inner diameter. The tabs have an interference fit to the liner lower press-fit diameter to hold the seal ring in place during liner installation.

Install the seal rings.

The seal ring **must** be straight on the liner upon installation. Use finger pressure to press the seal ring near the tabs to fit the seal ring down and over the lower press fit diameter during installation.

This practice during installation of the seal ring will prevent deformation that will result in the seal ring **not** fitting squarely on the bottom of the liner flange.



Cylinder Liner Seal Rings

Thickness	Number of Notches
0.457 mm [0.0180 in]	Two

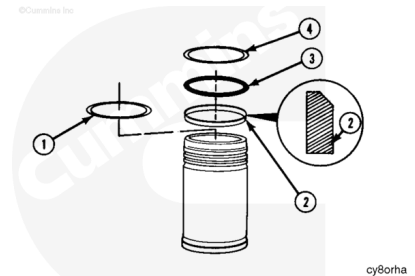
Cylinder Liner Seal Rings

Thickness	Number of Notches
0.521 mm [0.0205 in]	Two
0.559 mm [0.0220 in]	Two
0.610 mm [0.0240 in]	Two
0.787 mm [0.0310 in]	Two

Install the crevice seal. The beveled-edge of the crevice seal (2) **must** be positioned as shown.

Some o-rings have a "D" shape cross section. This type of o-ring **must** be installed with the flat side against the cylinder liner.

Install the o-rings in the position shown. Use the mold mark on the o-ring to check whether the o-ring is twisted.

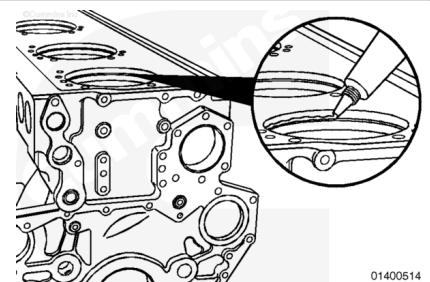


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1. Liner, counterbore sealing ring
2. Crevice seal
3. Black o-ring
4. Red o-ring.

⚠ CAUTION ⚠

If more than a 2 to 3 mm [0.078 to 0.118 in] bead of gasket sealant is applied to the cylinder block counter bore surface, engine damage can occur.



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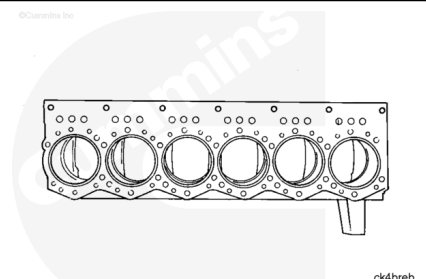
Install a bead of gasket sealant, Part Number 3165150, or equivalent, 2.0 to 3.0 mm [0.078 to 0.118 in] wide onto the cylinder block counter bore surface. The sealant bead **must** be a continuous loop and **must not** contain any breaks or voids from start to finish.

The assembly of the cylinder liner **must** be complete within 30 minutes of applying the gasket sealant.

⚠ CAUTION ⚠

Do not use petroleum-based oil. This will cause swelling of the packing ring.

Use vegetable oil to lubricate the inside diameter of the packing ring bores.



ck4breb

Note : There are two sized liners with different part numbers used in production and service. The liners are identified by labeling of "S" for small liners and "L" for large liners. "S" liners correspond with cylinders 1, 2, and 3. "L" liners correspond with cylinders 4, 5, and 6.

Use your hands to press the cylinder liners into the block.

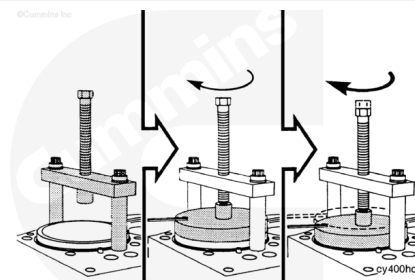
Use liner installation tool, Part Number 3164207, or equivalent, to install the liner. Install the bridge assembly and two cylinder head capscrews. Tighten the capscrews.

Torque Value: 68 n·m [50 ft-lb]

Install the pusher plate in the liner. Make sure it is aligned correctly in the liner. Turn the pusher screw until it touches the plate. Turn the pusher screw until the liner flange touches the counterbore ledge.

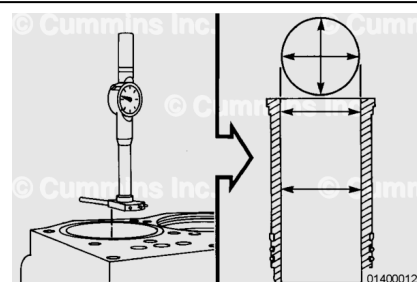
Tighten pusher screw.

Torque Value: 90 n·m [67 ft-lb]



Use a dial bore gauge and measure the inside diameter of the liner at the top, bottom, and middle of the liner.

Perform two measurements at each location. The measurements **must** be 90 degrees apart.



New Cylinder Liner Inside Diameter		
mm		in
170.000	MIN	6.693
170.040	MAX	6.694

If the inside diameter is more than 0.04 mm [0.002 in] out-of-round in the bottom measurement location, the liner **must** be removed. Check for a twisted o-ring.

Measure

Use liner installation tool, Part Number 3164207, or equivalent, and two cylinder head capscrews.

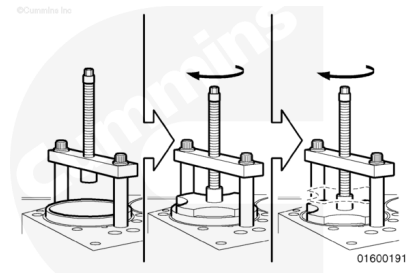
Tighten the capscrews.

Torque Value: 68 n·m [50 ft-lb]

Install the pusher plate in the liner. Make sure it is aligned correctly in the liner. Turn the pusher screw until it touches the plate. Turn the pusher screw until the liner flange touches the counterbore ledge.

Tighten the pusher screw.

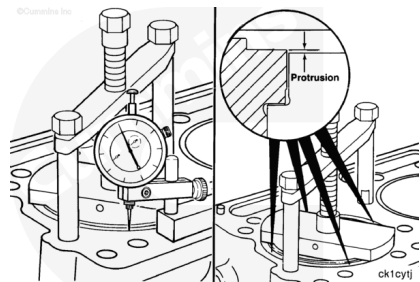
Torque Value: 90 n•m [66 ft-lb]



Use a depth gauge block. Position the depth gauge block so the indicator needle will contact the liner flange on the outside of the sealing bead as shown.

Gently press the indicator needle down until it touches the liner. Turn the gauge until the zero is aligned with the head of the dial. Repeat this step several times to be sure the gauge is set to zero.

Raise the indicator and move the gauge block until the indicator will touch the block surface. Gently press the indicator down until it touches the block. Read the indicator.

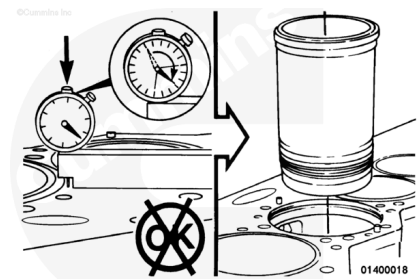


Cylinder Liner Protrusion		
mm		in
0.07	MIN	0.003
0.15	MAX	0.006

Read the liner protrusion in four places equally spaced on the liner outside diameter. The measurements on each liner **must not** vary more than 0.05 mm [0.002 in].

If the liner protrusion is **not** correct, the liner **must** be removed. The protrusion can be adjusted using seal rings and/or machining the counterbore ledge. Contact a Cummins® Authorized Repair Facility if machining is required.

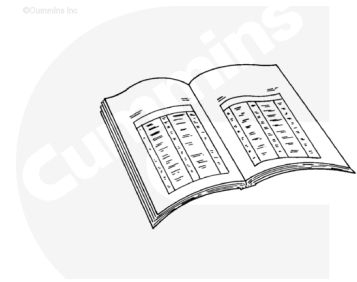
Note : Use seal rings to increase protrusion. Machine counterbore ledge to decrease protrusion.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the connecting rod and piston assembly. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html)
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Install the crossheads.
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Adjust the overhead set. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Install the air intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/89/89-011-007.html)
- Install the oil suction tube and the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/89/89-008-018.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Connect the batteries or air supply to the starter.
- Operate the engine. Check for leaks.

Last Modified: 17-Jan-2020
